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MLA0301

CO1-AT2- Case Study

Topic: A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping.

Title:

Design of a Reinforcement Learning Framework for Smart Home Automation using Occupancy and Weather Conditions.

1: Introduction

1.1 Introduction

Smart home automation is the process of automatically controlling household devices such as lighting, heating, ventilation, and air conditioning (HVAC). Traditional automation systems rely on fixed schedules or manually defined rules, which often fail to adapt to changing environmental conditions or user behavior.

Reinforcement Learning (RL) enables the smart home to learn optimal control strategies by interacting with the environment. An RL agent continuously observes occupancy, weather conditions, and indoor temperature, then decides the best actions to improve comfort while reducing energy consumption.

By receiving rewards based on its decisions, the agent gradually learns an energy-efficient policy that balances user satisfaction and operational cost.

1.2 Objectives

Design an RL framework for smart home automation.

Identify states, actions, and rewards.

Improve occupant comfort.

Reduce electricity consumption.

Analyze conflicting objectives.

Study the impact of reward shaping.

Figure 1: Smart Home Automation Overview

```
        Weather
          |
          ▼
      RL Smart Agent
      /    |    \
Lighting  Heating  Cooling
      \    |    /
   Home Environment
          |
          ▼
Comfort + Energy Saving
```

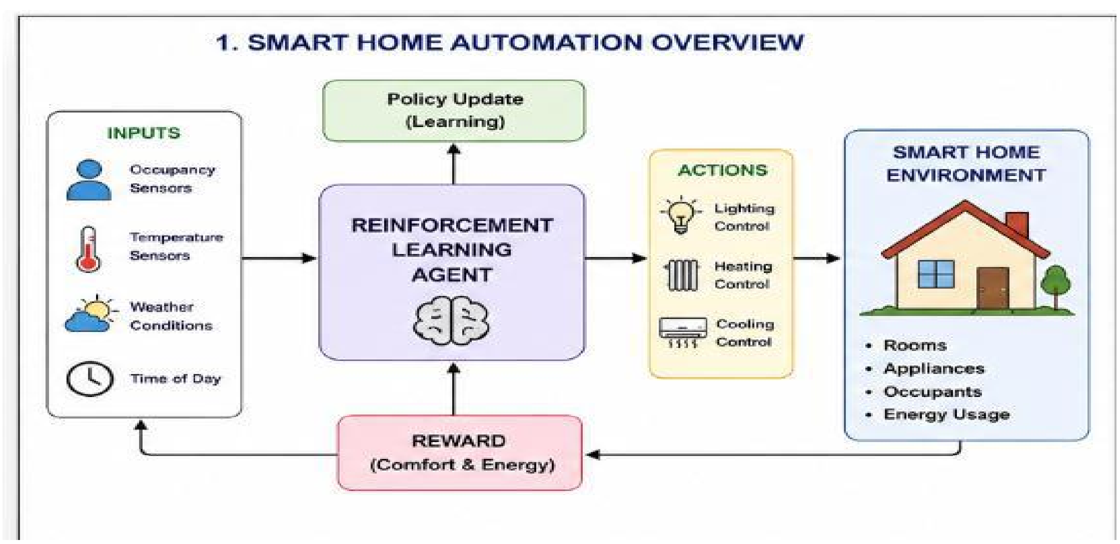


Figure 1: Overview

2: Reinforcement Learning Framework

Components

Agent

The intelligent controller that decides how to operate the appliances.

Examples:

Turn lights ON/OFF

Increase room temperature

Reduce cooling

Environment

The smart home itself including

Rooms

Occupants

Temperature

Weather

Electrical appliances

States

The state represents the current condition of the house.

Example state variables

Occupancy

Indoor temperature

Outdoor temperature

Weather

Time of day

Energy usage

Actions

The agent can perform actions such as

Turn light ON

Turn light OFF

Increase heater

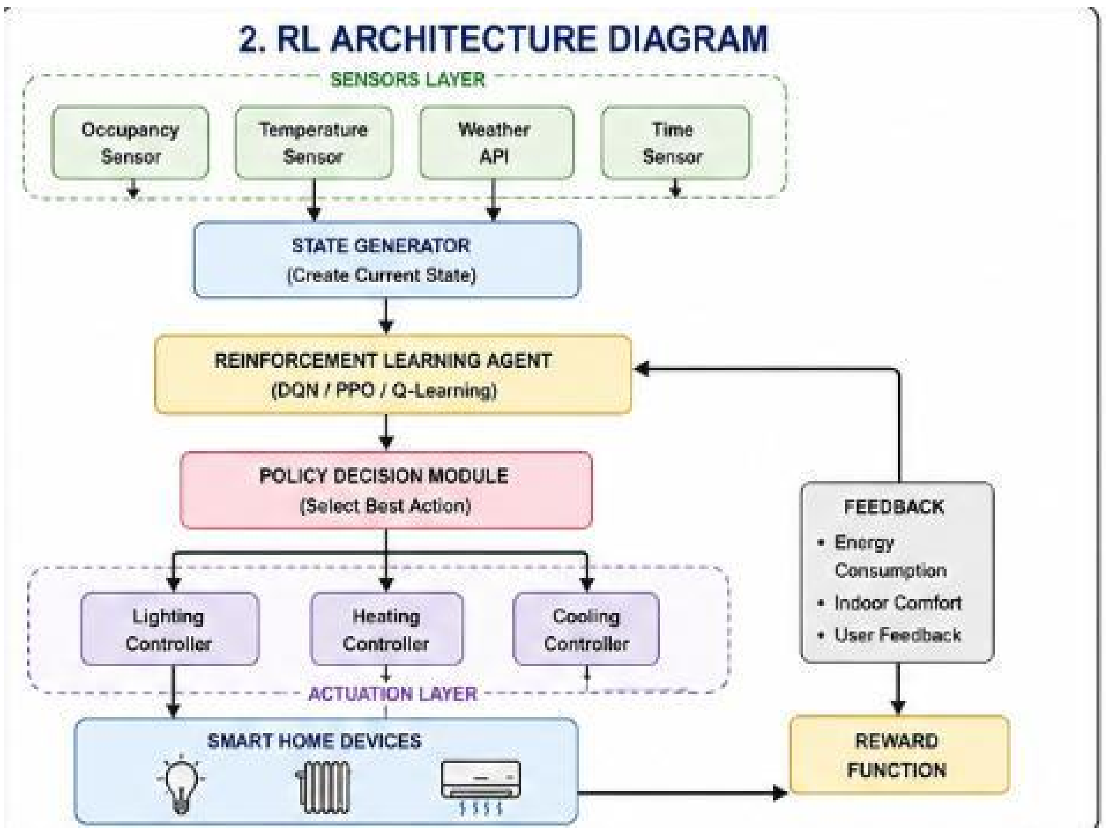
Decrease heater

Increase cooling

Decrease cooling

Maintain current settings

Figure: RL architecture diagram



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3.Rewards

The environment provides rewards based on the chosen action.

Positive reward

Comfort maintained

Energy reduced

+10

Negative reward

Room too hot

Lights ON in empty room

-8

RL Interaction Diagram

```
        Current State
              |
              ▼
         RL Agent
              |
        Select Action
              ▼
   Smart Home Environment
              |
     Reward + Next State
              ▲
              |
```

Repeat Learning

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Workflow

1. Sensors collect data.

2. State is generated.

3. RL agent observes state.

4. Action is selected.

5. Devices operate.

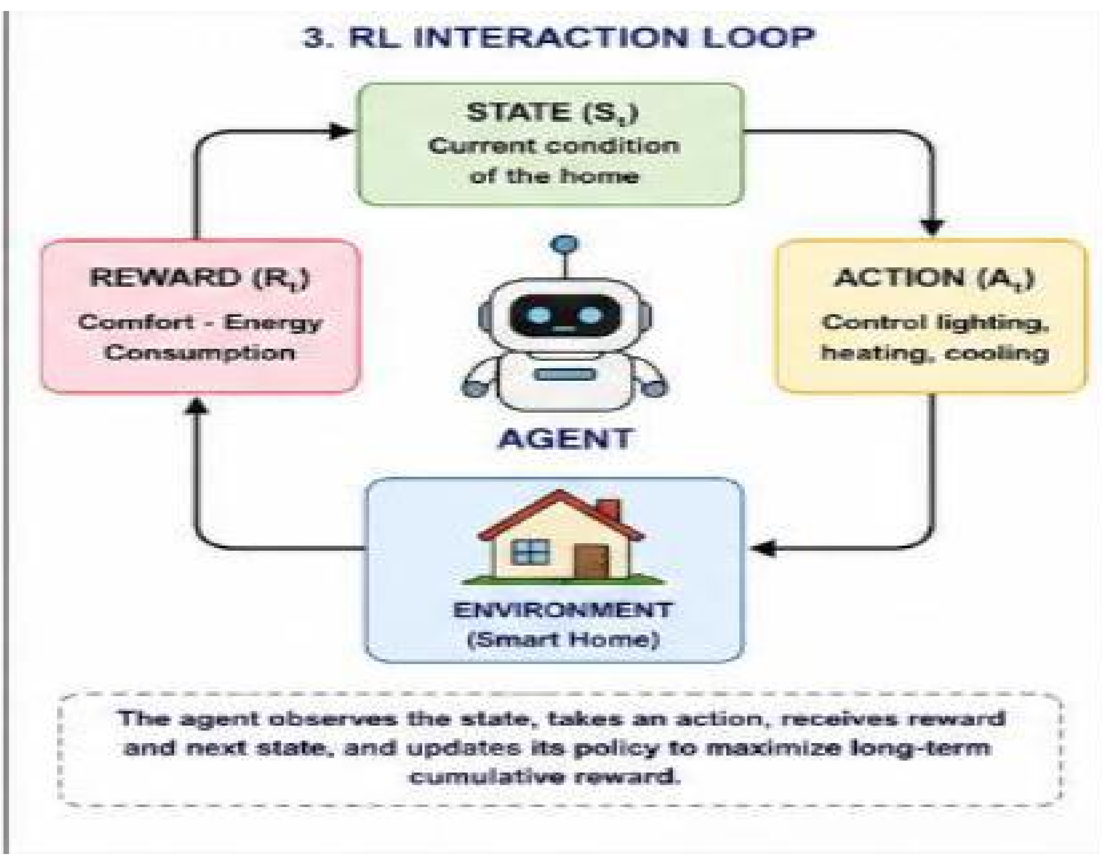
6. Energy and comfort are measured.

7. Reward is calculated.

8. Agent updates policy.

Figure 3: Interaction loop

4.Reward Function



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A simple reward function can be written as:

Reward = Comfort Score − Energy Consumption

Where:

- Comfort Score increases when indoor conditions match user preferences.

- Energy Consumption decreases the reward when excessive electricity is used.

This encourages the agent to maximize comfort while minimizing power usage.

5: Conflicting Objectives

The biggest challenge is balancing comfort and energy saving.

Scenario 1

Winter

Occupant is inside.

Desired temperature = 24°C

The RL agent increases heating.

Comfort is high.

Energy consumption also increases.

Scenario 2

Nobody is home.

The RL agent turns OFF lights.

Heating is reduced.

Energy saving increases.

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Conflict Table

Situation

Comfort

Energy Saving RL Decision

Occupied room High

Medium

Maintain comfort

Empty room

Low priority High

Turn devices OFF

Hot weather

Medium

Medium

Moderate cooling

Cold weather

High

Medium

Moderate heating

Handling Conflicts

The RL agent learns priorities through the reward function:

•

Positive rewards for maintaining comfortable conditions.

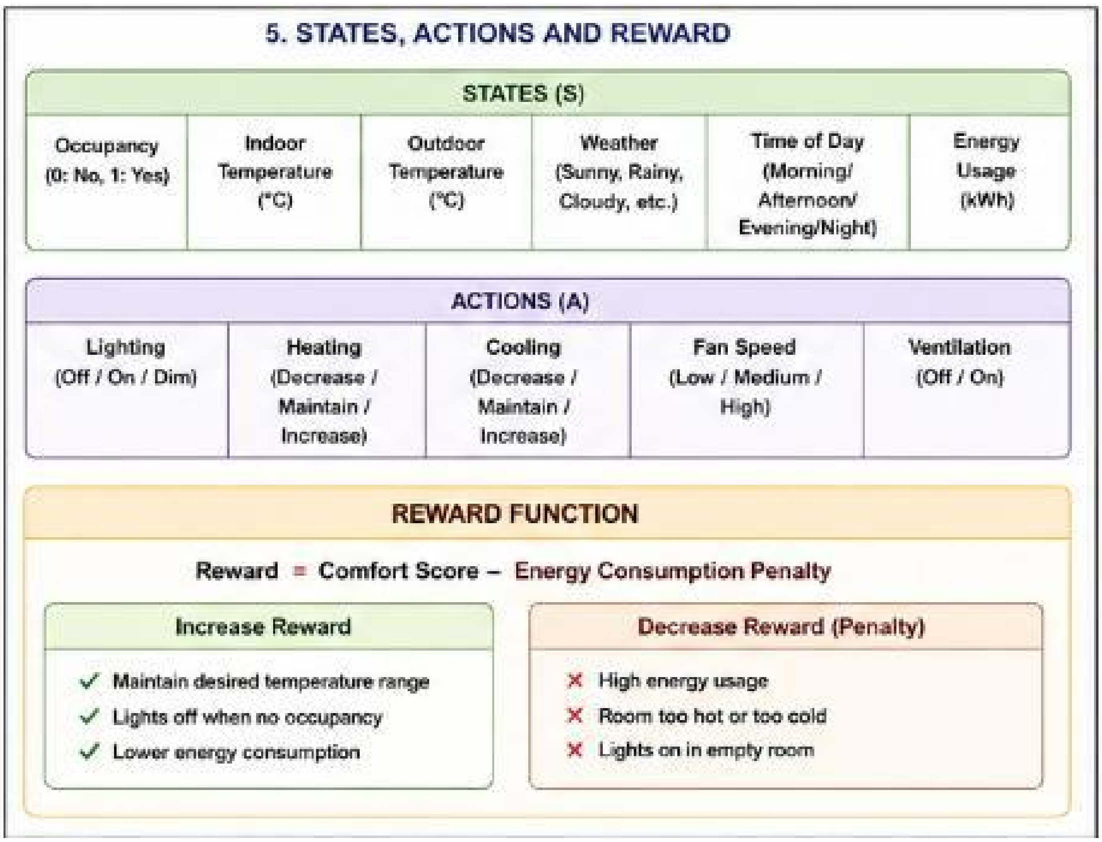
•

Negative rewards for unnecessary energy usage.

•

Balanced decisions based on occupancy and weather.

Figure 4: States , Actions and Reward



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6: Impact of Reward Shaping

Reward shaping provides additional guidance to help the RL agent learn faster.

Without Reward Shaping

- Learning is slower.
- The agent explores many poor actions.
- Higher energy wastage during training.

With Reward Shaping

- Faster convergence.
- Improved comfort.
- Reduced energy consumption.
- More stable decision making.

Comparison Table

Parameter

Without Reward Shaping With Reward Shaping

Learning Speed

Slow

Fast

Energy Saving

Medium

High

Occupant Comfort Moderate

High

Policy Stability

Low

High

Advantages

•

Lower electricity bills.

•

Better indoor comfort.

•

Intelligent adaptation to weather.

Reduced human intervention.

•

Sustainable energy usage.

Page 7: Conclusion & Future Scope

Conclusion

This case study demonstrated the design of a Reinforcement Learning framework for smart home automation. The RL agent learns from occupancy, weather, and temperature data to control lighting and HVAC systems efficiently. By carefully designing states, actions, and reward functions, the system balances user comfort with energy efficiency. Reward shaping further improves learning speed and overall performance.

Future Scope

- Integration with solar energy systems.
- Voice-controlled automation.
- Multi-room optimization.
- Deep Reinforcement Learning (DQN, PPO).
- Smart grid integration.
- IoT-enabled predictive control.